

Niriantsoa Francia RAHERINIAINA

M.Sc. Candidate in Astronomy & Astrophysics

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Summary

Master's student in Astronomy and Astrophysics with specialized training in radio astronomy, VLBI data reduction, and scientific computing through the DARA programme. Experienced in astronomical data analysis, Python programming, and observational astronomy. Eager to apply machine learning to radio astronomy through hands-on experience with MeerKAT data.

Education

University of Antananarivo

Master 2 (M.Sc.) in Astronomy & Astrophysics

Antananarivo, Madagascar

Ongoing

- Relevant coursework: Galaxies, Cosmology, Python Programming, Linux.

DARA Basic Training Programme

Development in Africa with Radio Astronomy

2025 - 2026

- Linux & Python for Scientific Computing
- Optical & Radio Astronomy
- VLBI and Radio Astronomy Data Reduction
- Scientific Collaboration & Professional Development

École Normale Supérieure (ENS)

Bachelor's Degree in Physics & Chemistry Education

Antananarivo, Madagascar

2021 - 2024

- Graduated with Honours (14/20)

Selected Projects

Moon Phase Calculator (*Python, Astropy, PySide6*)

[GitHub](#)

- Developed a Python application using **Astropy** to calculate lunar phases and illumination.

Internet Movie Database (IMDb) movie dataset analysis (*Python, Pandas, Matplotlib*)

[GitHub](#)

- Applied Extract, Transform, Load (ETL) and Exploratory Data Analysis (EDA) techniques on an IMDb movie dataset to answer key analytical questions.

Technical Skills

- Programming: Python, NumPy, Pandas, Matplotlib, Astropy, Linux, Git/GitHub
- Astronomy: CASA, SAOImage DS9, Astrometrica, Radio Interferometry, VLBI, FITS Data Analysis
- Data Analysis: Scientific Computing, Data Visualization, Data Pre-processing

Research Interests

Radio Astronomy • Machine Learning • Artificial Intelligence • Scientific Computing • Signal Processing • Astronomical Data Analysis

Conferences

- DARA Annual Network Meeting, South Africa (2026)
- TART Conference, Madagascar (2026)

Languages

- Malagasy : Native
- French : Fluent
- English : Professional Working Proficiency

References

- **Prof. Solohery Randriamampandry**
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- **Dr. Andry Rasoanaivo**
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